

INDUSTRIAL BRAIN	Inspection Report	Doc ID: Inspection_Report_Dec2023
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Post-Incident Teardown Report

Date: 2023-12-03 **Inspector:** Alex Chen, Reliability Engineer **Asset:** Pump P-101

Teardown Findings

- **Outboard Bearing:** Completely shattered. Cage is destroyed, rollers are deformed. Heavy bluing on the metal indicating extreme heat (lack of lubrication).
- **Shaft:** Scored and bent due to bearing seizure. Requires replacement.
- **Impeller:** Minor rubbing on the casing wear rings.

Root Cause Analysis

The bearing failed due to lubrication starvation. A review of the Q3 Maintenance Log shows that the quarterly lubrication (mandated by SOP-01 and the OEM Manual) was skipped in August due to a lack of SynthoLube-X9 grease. The bearing ran dry for over 6 months, leading to the elevated vibrations seen in Nov 12th Inspection, and ultimate catastrophic seizure on Dec 2nd. The seizure of the pump caused a mechanical overload on Motor M-22, which correctly tripped its thermal protection relay.

Prepared By / Technician

Approved By / Supervisor